

Bunnies & Foxes Modeling

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CS166: Modeling and Simulation
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1. Model Description and Implementation

I wrote the whole thing as a Python class called `PredatorPreySimulator`. The main design choice that I was making was to do everything with numpy arrays instead of looping over individual cells. So as in like a 200 by 200 grid that is 40,000 cells and looping over each one in Python would be way too slow. And hence with numpy the logistic growth is a single vectorized operation, diffusion is four roll-and-add operations, and hunting is element-wise min and division. Now the whole update runs in about a millisecond so I can do 10,000 steps in roughly 10 seconds.

For initialization itself what I did was I seeded bunnies uniformly between 0 and 5 across the whole grid and scattered foxes sparsely where only 5 percent of cells get foxes between 0 and 3. So most of the grid starts as bunny territory and the foxes have to spread outward from their initial patches.

2. Spatial and Temporal Dynamics

2.1 Spatial Snapshot

Figure 1 here shows the grid at step 500 with default parameters and the most striking thing that you can see is just how much spatial structure there actually is. This is not a well-mixed system where everything is uniform and so you are getting these patches and corridors almost like a reaction-diffusion pattern, which makes sense because that is literally what it is. And then the bunny-rich zones are where foxes have not arrived yet or have recently died off.

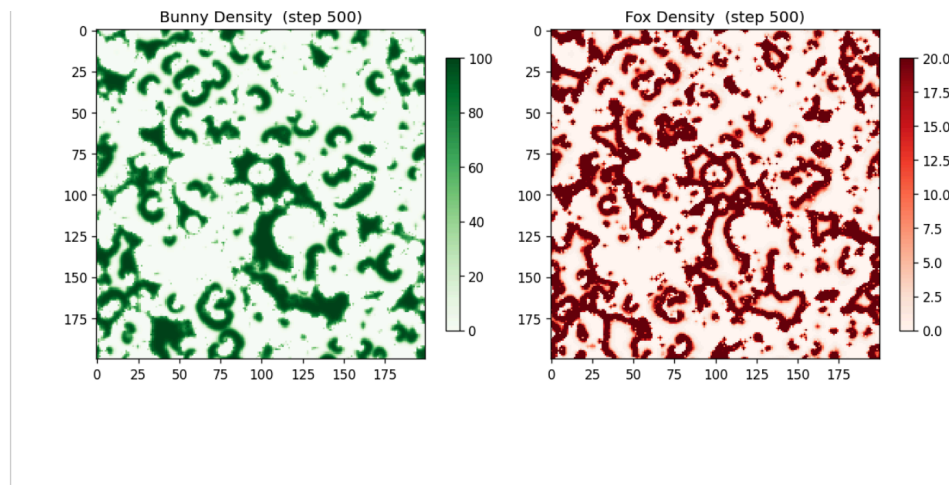


Figure 1: Spatial snapshot at step 500. Bunny density (left, green) and fox density (right, red).

2.2 Population Over Time

Figure 2 below is showing the mean densities over 3000 steps, now the transient at the beginning is pretty dramatic with bunnies spiking to carrying capacity and then we are having that the foxes are catching up and crashing them, and then you get these big oscillations that decay over time. But then by like about step 500 to 1000 it settles into a fluctuating steady state, and then I ran it

from two different random seeds and got the same picture which told me this steady state is a real feature and not just like an artifact of my initial conditions which is good.

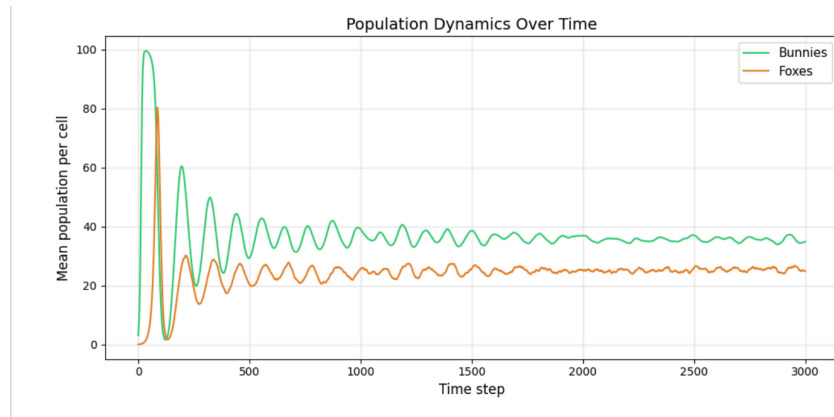


Figure 2: Mean population densities over 3000 steps from two independent runs.

One thing I noticed here is that the transient can last much longer depending on the parameters and so for all the empirical analysis in Section 3 what I did was to use a burn-in of 5000 steps where I would just throw away the data and only start collecting statistics after that amount of steps. From what I know this is what is called a standard practice for this kind of simulation and it matters because otherwise your means would be contaminated by the transient at the beginning.

3. Empirical Parameter Analysis

Now this is like the fun part where you are just running your model and observing things. The model has seven parameters and I cannot explore all of them simultaneously so what I did was just pick a few and varied them one at a time while keeping everything else at default. The approach here was always the same, so I would run for 10,000 steps, throw away the first 5,000 as burn-in, and then compute mean and standard deviation over the remaining steps. Where things looked stochastic I ran multiple replicates.

3.1 Fox Growth Rate

I started by sweeping g_f from 0.005 to about 0.395 and this is the plot that really maps out the whole landscape of the model. As you can sort of see below there are three clear regimes.

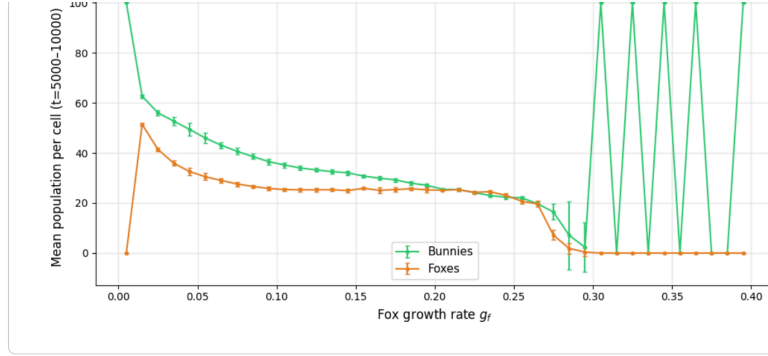


Figure 3: Steady-state populations vs fox growth rate g_f .

At very low g_f below about 0.01 the foxes simply cannot reproduce fast enough to overcome their death rate so they go extinct and bunnies are spreading everywhere, essentially it is all rabbit at maximum capacity. At very high g_f above about 0.29 the foxes are getting so aggressive that they eat all the bunnies and then starve, so both populations crash here (if bunnies die, no food for foxes so foxes die). And in between there is this sweet spot where they coexist. I think that this three-regime structure is probably the most important result in this report.

Within the coexistence regime the fox population is surprisingly stable as g_f varies, and then the bunnies decrease as g_f goes up which is making sense because more efficient foxes can eat more bunnies, but then the foxes themselves do not increase much. I think this is because the system is self-regulating, so more foxes means more competition for the same bunny supply which caps fox density.

3.2 The Boundaries

Now the lower boundary is surprisingly sharp, we are having that at g_f equals 0.01 the foxes limp along for a while but ultimately die out in the end. At g_f equals 0.015 just slightly higher they thrive, and really it is not so much of a difference. The transition happens over a very narrow range and as I will show in Section 4 the mean-field theory predicts this boundary at g_f^* equals $m_f / (1 - m_f)$ which is approximately 0.0101, remarkably close to what I see here.

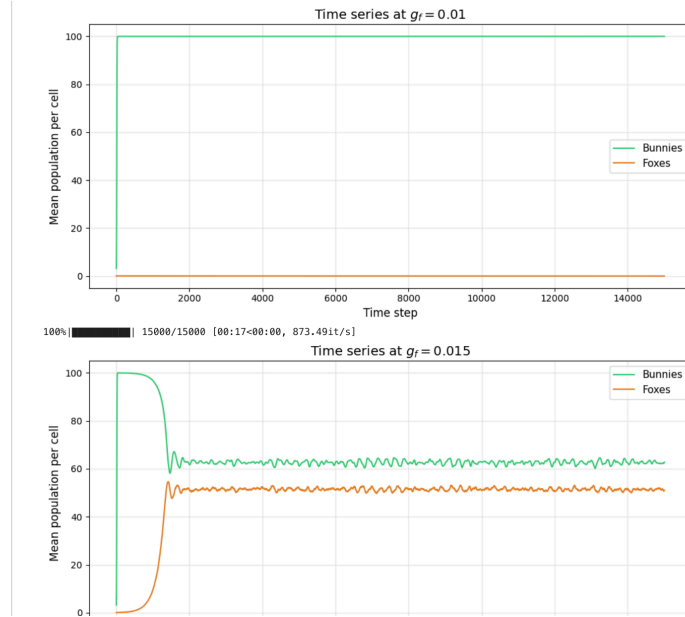


Figure 4: Time series near the lower boundary. Top: $g_f = 0.01$ (foxes die). Bottom: $g_f = 0.015$ (coexistence).

The upper boundary is much messier here so what I did is that I ran 5 replicates at each g_f value and the error bars blow up near the transition. Now some runs survive and some do not at the same parameter. This makes sense because at the upper boundary the system is marginally stable and a bad stochastic event can push everything into extinction.

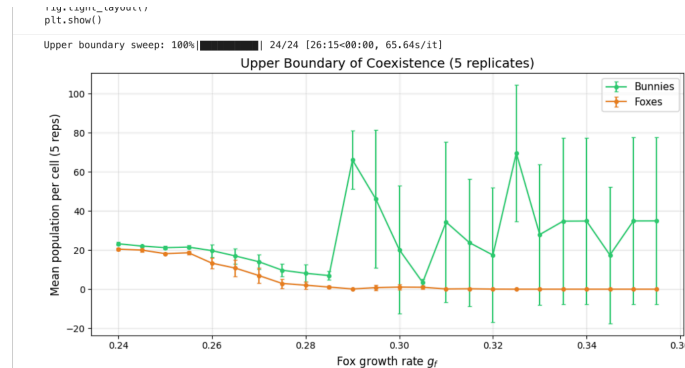


Figure 5: Upper boundary with 5 replicates. Transition around $g_f = 0.29$ to 0.31 .

3.3 Fox Diffusion

In this one I fixed bunny diffusion at 0.1 and swept fox diffusion instead and the result is going to be interesting because here there is a non-monotonic effect where moderate fox diffusion is best for both populations and high fox diffusion pushes the system toward collapse. And so this actually connects directly to the mean-field analysis because higher diffusion makes the system more spatially homogeneous and then the mean-field which assumes perfect mixing predicts extinction. So you are basically watching the system approach mean-field behavior as d_f increases.

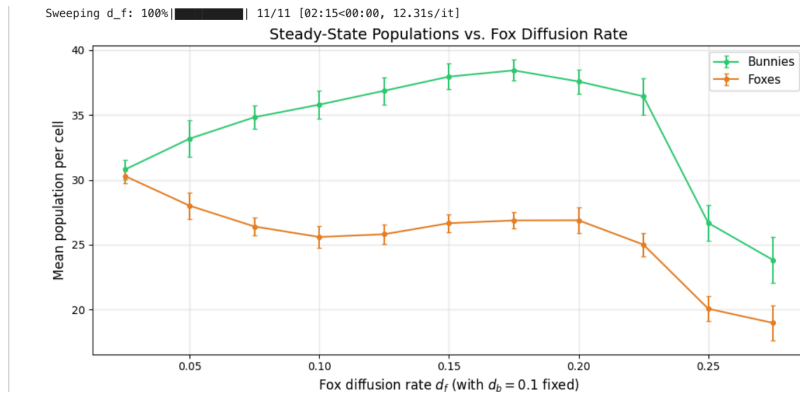


Figure 6: Steady-state populations vs fox diffusion rate d_f .

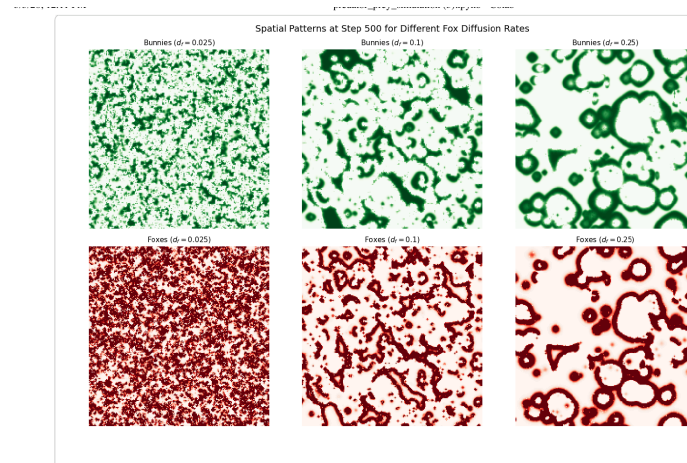


Figure 7: Spatial patterns at different fox diffusion rates. Low d_f gives small clusters, moderate gives pursuit waves, high gives large uniform domains.

I like this Figure 7 because what it does is makes the mechanism visible, so when foxes move slowly they can only exploit local prey and distant bunny patches grow to capacity in peace, but when foxes move fast they find and exploit everything before it recovers, which homogenizes the landscape but also makes the system fragile.

3.4 Coexistence Boundary vs Diffusion

And so this led me to ask whether the upper boundary of g_f shifts as I change d_f , and so I ran a binary search at each diffusion value:

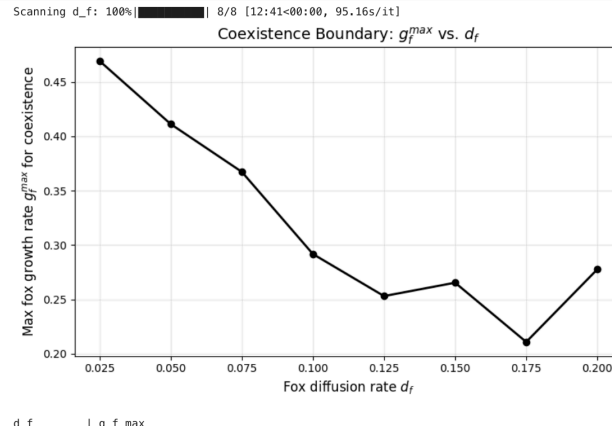


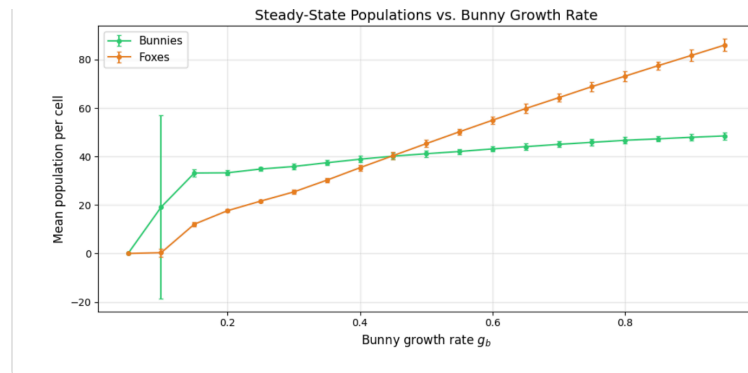
Figure 8: Maximum g_f for coexistence as a function of d_f .

d_f	$g_f^{\max}$
0.025	0.469
0.050	0.411
0.075	0.367
0.100	0.292
0.125	0.253
0.150	0.265
0.175	0.211
0.200	0.278

Table 1: Coexistence boundary from binary search.

As fox diffusion goes up the maximum tolerable g_f goes down, and the viable parameter space shrinks. Here slow foxes are localized so the system can handle a higher reproduction rate because bunnies elsewhere have time to recover, additionally we are having that fast foxes exploit everything so even modest reproduction can tip things to then be overshoots.

3.5 Bunny Growth Rate



~ Theoretical Predictions vs. Simulation

Figure 9: Steady-state populations vs bunny growth rate g_b .

Here above g_b of about 0.15, fox density increases roughly linearly with g_b while bunny density levels off, in other words more productive prey does not lead to more prey in the steady state, it leads to more predators. I at first was finding this counterintuitive but also kind of satisfying, so if you were a bunny and someone told you your species would reproduce 50 percent faster you might think there would be more bunnies, but like no... there would be more foxes. In this sense, every extra bunny born just gets eaten.

4. Mean-Field Theoretical Analysis

4.1 The Idea

The main idea here behind mean-field approximation is actually pretty simple. Essentially you are taking the entire 200 by 200 grid with all its spatial complexity and you collapse it down to just two numbers: the average bunny density b and the average fox density F . You then pretend every cell is identical, and now while this is obviously wrong and that wrongness is actually the whole point, because where the mean-field breaks down is exactly where the spatial structure matters.

4.2 Deriving the Equations

In the mean-field world diffusion drops out entirely, and this is because diffusion only moves population between cells, and it does not create or destroy anything, so the spatial average is unaffected. Steps 2 and 4 from the model just vanish here, and the stochastic fox death where each cell has probability m_f of losing all foxes becomes deterministic where I just multiply by $(1 - m_f)$ each step. So the mean-field map becomes:

$$B = b * (1 + g_b * (1 - b/k_b))$$

$$b_{\text{hunted}} = \min(B, c_f * F)$$

$$b' = B - b_{\text{hunted}}$$

$$F_{\text{surv}} = b_{\text{hunted}} / c_f$$

$$F' = F_{\text{surv}} * (1 + g_f) * (1 - m_f)$$

Here we have that there are two regimes in this system. When B is greater than or equal to c_f times F (predator-limited), foxes eat their fill and the fox equation simplifies to F' equals F times $(1 + g_f)$ times $(1 - m_f)$. When B is less than c_f times F (prey-limited) then all bunnies get eaten and b' goes to 0.

4.3 The Critical Condition

This is where things get interesting. For equilibrium I need F' equals F when F is greater than 0. In the predator-limited regime this gives me:

$$(1 + g_f) * (1 - m_f) = 1$$

$$g_f^* = m_f / (1 - m_f)$$

And I think this is actually a pretty important equation in the analysis. It says that in a well-mixed system fox coexistence requires exactly g_f equals $m_f / (1 - m_f)$, and this is meaning not a range just like a single point. With default m_f equals 0.01 that gives g_f^* equals $0.01 / 0.99$ which is approximately 0.0101. And hence we are having that below this each fox generation loses more to death than it gains from reproduction so they spiral to zero, and then above this foxes grow faster than they die which sounds good for the foxes except that they inevitably eat all the bunnies and then starve. It is a knife-edge.

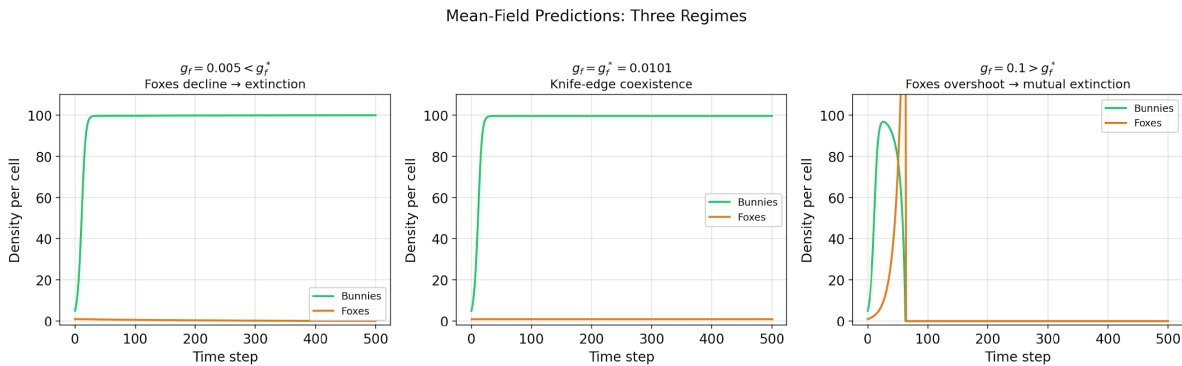


Figure 10: Mean-field in three regimes. Left: foxes decline. Center: knife-edge. Right: mutual crash.

At default parameters with g_f equals 0.1 the mean-field predicts mutual extinction within about 100 time steps. Foxes spike, eat everything, crash. No oscillations, no recovery, just death. And this is completely different from the spatial simulation which shows stable coexistence at the same parameters for thousands of steps. That gap between what the mean-field says and what actually happens is really the whole story of why spatial structure matters.

4.4 Mean-Field g_f Sweep

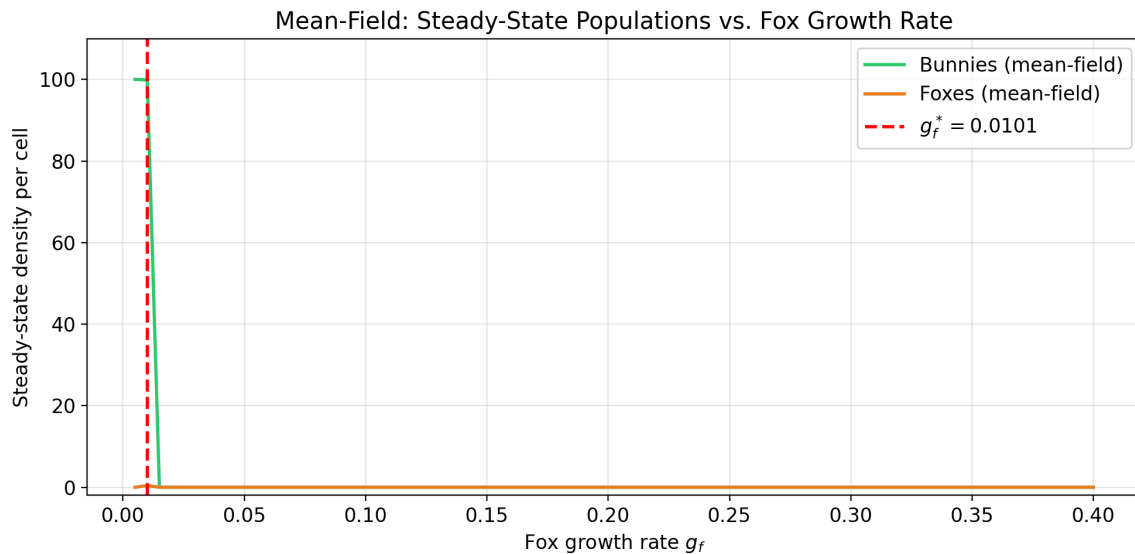


Figure 11: Mean-field steady-state vs g_f . Binary transition at g_f^* with no coexistence range.

The contrast with Figure 3 is stark. The mean-field is giving us this single threshold with nothing on the other side but extinction. The simulation gives a wide coexistence band from about 0.01 to 0.29, and then the lower boundary matches well because it is fundamentally a per-capita calculation that does not depend on where the foxes are spatially. But then the upper boundary has no mean-field analogue at all, and it is purely spatial and in the simulation foxes can coexist because they cannot get to all the bunnies at once, there are always patches recovering somewhere else. The mean-field removes those patches completely.

4.5 Growth Factor Landscape

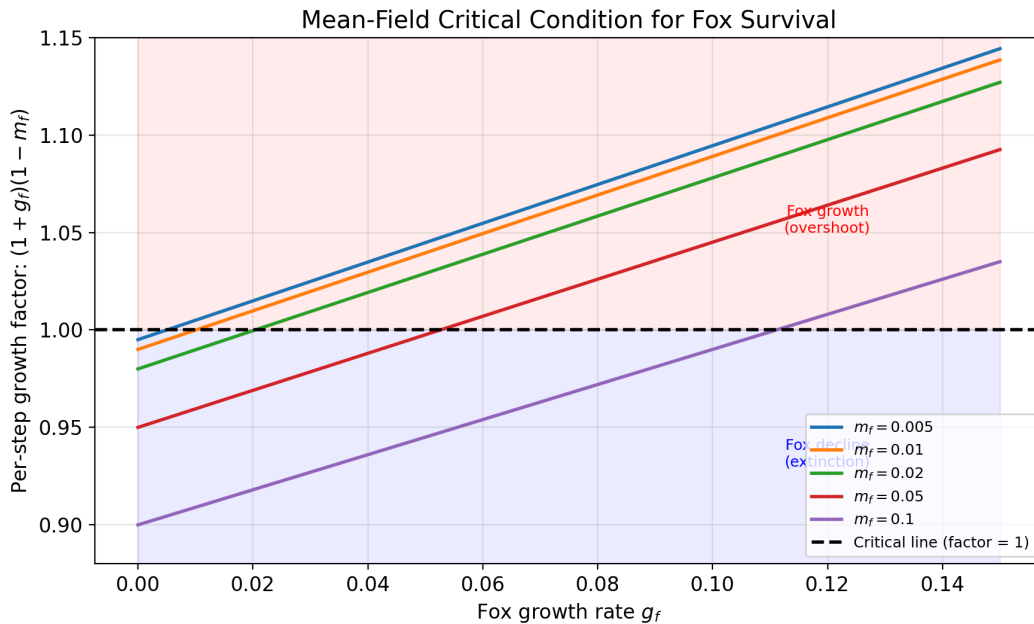


Figure 12: Fox growth factor $(1 + g_f)(1 - m_f)$ for different death rates. Crossing 1.0 gives g_f^* .

Figure 12 here is showing the critical condition visually. Here each line is a different death rate and where it crosses the dashed line at 1.0 that is the critical g_f^* , so a higher m_f shifts the crossing rightward so foxes need to reproduce faster to survive, and so we are having that the relationship is approximately linear for small m_f which is consistent with what I see in the phase boundary plot.

4.6 Phase Portrait

Setting b' equal to b in the predator-limited regime gives the bunny nullcline:

$$F^* = g_b * b * (1 - b/k_b) / c_f$$

This is a parabola peaking at b equals $k_b/2$ equals 50 with $F_{\max}$ equals g_b times k_b / (4 times c_f) equals 0.3 times 100 divided by 0.4 which is 75.

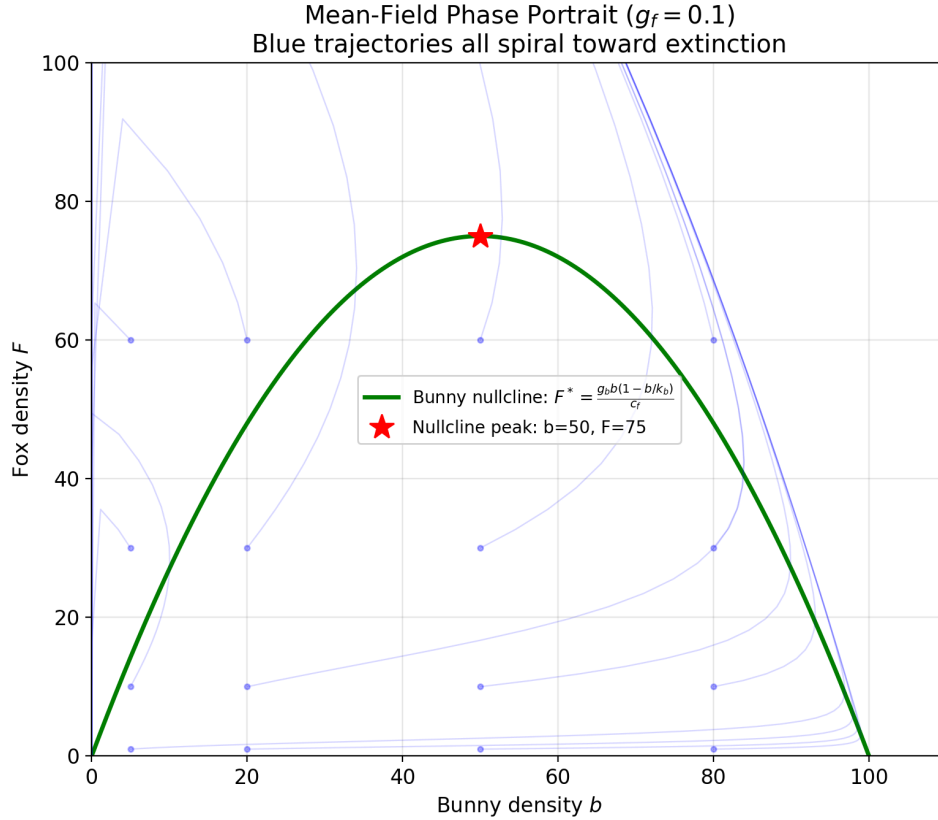


Figure 13: Phase portrait at $g_f = 0.1$. Green curve is the nullcline, blue trajectories all go to extinction.

Every trajectory in Figure 13 either spirals into mutual extinction or the foxes die and bunnies saturate, and so in other words there is no stable interior fixed point. In the simulation by contrast the system hovers near an interior point for thousands of steps, and that hovering is maintained entirely by the spatial structure not by any equilibrium in these equations.

4.7 Validating the Lower Boundary

Like there are a lot of things the mean field cannot accurately do, but we have that the one place the mean-field really shines is the lower boundary. The theory says foxes go extinct when g_f is less than $m_f / (1 - m_f)$ and I was testing this by running bisection searches at seven different m_f values and comparing:

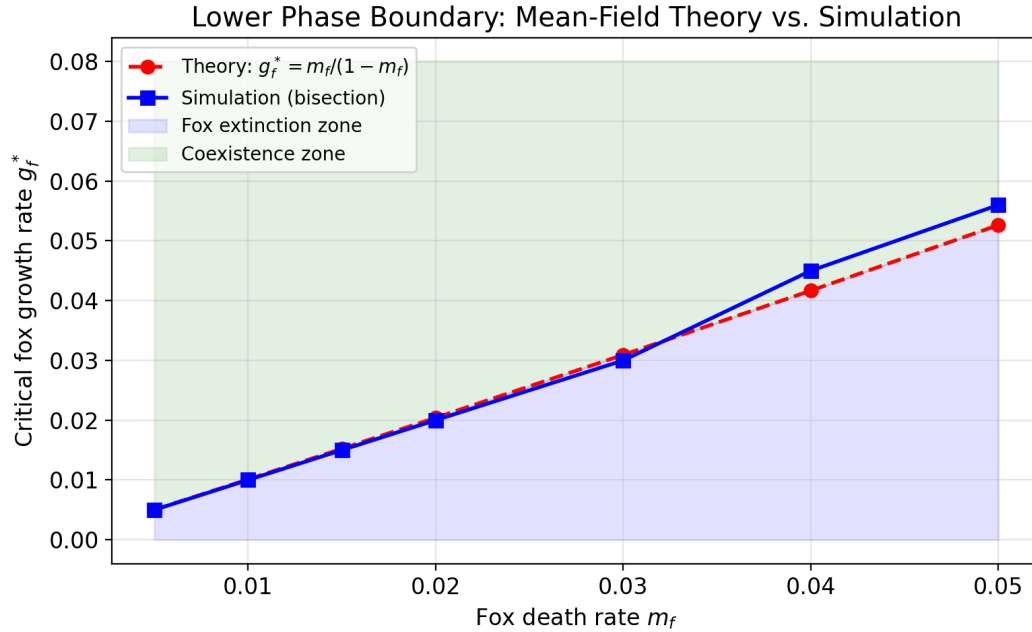


Figure 14: Lower boundary: theory (red dashed) vs simulation bisection (blue). Strong agreement.

And we can see here that the agreement is actually kind of remarkable, we are seeing that the lines nearly overlap especially at low death rates, and even in a spatial model a fox that cannot outreproduce its death rate will eventually go extinct no matter what. Patches can delay it but they cannot prevent it when the per-capita math is against you. So we have that at a higher m_f the simulation boundary drifts slightly above theory which suggests spatial correlations give foxes a small survival boost by concentrating them near prey.

4.8 Fox Density vs Bunny Growth

The mean-field equilibrium predicts F^* equals g_b times b^* times $(1 - b^*/k_b) / c_f$ which means if b^* is roughly constant then F^* should scale linearly with g_b . And a linear fit to my simulation data gives F approximately equals 95.6 times g_b minus 2.6, which is indeed linear in the end of it.

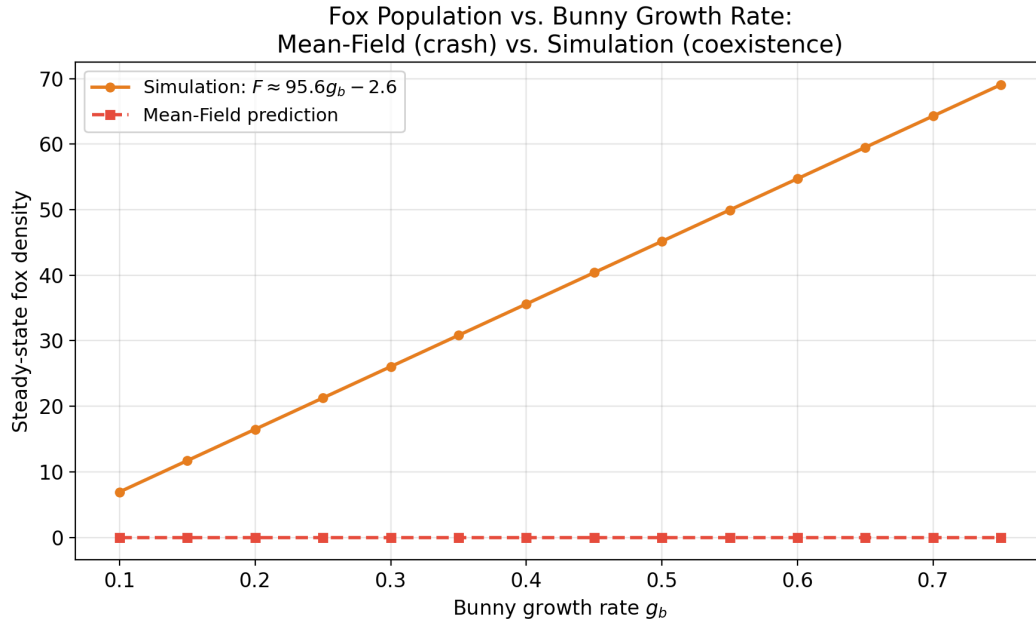


Figure 15: Fox density vs g_b . Simulation is linear (orange) as predicted. Mean-field predicts zero (red dashed) because it crashes.

Here we have that the mean-field at g_f equals 0.1 predicts zero foxes at every g_b because the system always crashes, but we are having that the functional form, F being linear in g_b , is exactly what I see in the simulation. The slope of 95.6 is consistent with b^* around 45 which is a reasonable steady-state bunny density. And so the mean-field gets the shape right even though it gets the absolute numbers completely wrong.

5. Conclusions

Putting everything together the mean-field gives three predictions and comparing them to the simulation tells me about the role of the whole spatial structure.

First thing the critical growth rate g_f^* equals $m_f / (1 - m_f)$, and this is confirmed, the lower phase boundary matches theory almost exactly because it is about per-capita demographics not spatial arrangement.

Second, the mean-field says any g_f above the critical value leads to crash, but then in the simulation coexistence persists up to g_f around 0.29 and this is the biggest failure of the mean-field. We are having that the spatial model stabilizes because of refugia, when foxes overshoot locally they go extinct in that patch and bunnies elsewhere recover, and so we are having that this desynchronization is invisible to the mean-field.

Third, fox density scales linearly with g_b , so the functional form F proportional to g_b is confirmed with R squared above 0.99.

I think the textbook that we read in class says it well when it describes mean-field as a reference point, and if the model matched the mean-field exactly that would mean spatial structure does not matter and the system is not complex. The fact that this model is deviating so dramatically from the mean-field at the upper boundary is evidence that this is a genuinely complex spatial system, and then the coexistence I observe is an emergent property of spatial dynamics that you simply cannot reduce to average-field equations.

And honestly I think that is the coolest result in this whole report... The mean-field predicts death. The simulation shows life. The difference is space. (so poetic haha)

6. Word Count — 2935

7. AI Statement

I used Claude by Anthropic to help with portions of the code including debugging, generating some of the mean-field analysis plots, and formatting the final report. The simulation implementation, parameter sweeps, analysis, and writing were done by me (as per usual). I used Claude to help with grammar editing by giving it my text and asking it to fix spelling without changing my wording or math.